Chapter 20 Notes

Section 2

Notes

Note that, in Definition 2.2, θ is simply being extended; so that $\theta(p)$ is still defined.

It is remarked in Definition 2.1 that, if θ is an automorphism, then so is θ^{-1}. This is because $\theta^{-1}(p)\mathcal{R}\theta^{-1}(q)$ iff $f(g) = p$ and $f(h) = q$ for some g and q, meaning $f(g)\mathcal{R}f(q)$ iff $g\mathcal{R}q$ (because f is an automorphism).

It is additionaly remarked that, if ϕ and ψ are automorphisms, then so is $\phi\psi$. This follows because $fg(p)\mathcal{R}fg(q)$ iff $g(p)\mathcal{R}g(q)$ iff $p\mathcal{R}q$.

Note that, if θ is a function obtained by Definition 2.2, then θ^{-1} is also such a function. That is, $\theta^{-1}(f) = \{\langle\theta^{-1}(p), \theta^{-1}(g)\rangle | \langle p, g\rangle \in f\}$ for all $f \in \mathcal{D}^{\mathcal{G}}$ (and the function is still 1-1 and the domain and ranges stay the same). This can be shown in the following way. $\theta^{-1}(f)$ is some h such that $\{\langle\theta(p), \theta(g)\rangle | \langle p, g\rangle \in h\} = f$. Then

$$\begin{array}{rl} & \{\langle\theta^{-1}(p), \theta^{-1}(g)\rangle | \langle p, g\rangle \in f\} \\ = & \{\langle\theta^{-1}\theta(p), \theta^{-1}\theta(g)\rangle | \langle p, g\rangle \in h\} \\ = & \{\langle p, g\rangle | \langle p, g\rangle \in h\} \\ = & h \\ = & \theta^{-1}(f) \end{array}$$

Also, if ϕ and ψ are functions obtained by Definition 2.2, then $\phi\psi$ is also such a function. That is, $\phi\psi(f) = \{\langle\phi\psi(p), \phi\psi(g)\rangle | \langle p, g\rangle \in f\}$ for all $f \in \mathcal{D}^{\mathcal{G}}$ (this just follows by the definition of ϕ, ψ, and function composition). The fact that $\phi\psi$ is still 1-1 and has a domain and range of $\mathcal{G} \cup \mathcal{D}^{\mathcal{G}}$ is also easily seen.

I did not fully understand the paragraph titled "**Notation**" on pg. 269. In particular, it was not fully clear to me why the recursion taking place in that paragraph was possible.

Exercise 2.1

We prove Proposition 2.3. First, we show tha $\theta : \mathcal{D}^{\mathcal{G}} \xrightarrow[\text{onto}]{\text{1-1}} \mathcal{D}^{\mathcal{G}}$. By definition, the domain of θ includes all members of $\mathcal{D}^{\mathcal{G}}$. We use transfinite induction to show that θ is 1-1, and is such that the range of θ includes all members and only members of $\mathcal{D}^{\mathcal{G}}$.

Proof. The least ordinal case is obvious.

Suppose the result holds for some α. That is, for every $f \in \mathcal{R}^{\mathcal{G}}_{\alpha}$, the following three things hold: (1) for all $g \in \mathcal{D}^{\mathcal{G}}$, if $\theta(f) = \theta(g)$, then $f = g$ (i.e., that θ is 1-1 for all members of $\mathcal{D}^{\mathcal{G}}$); (2) $f \in \text{Ran}(\theta)$; and (3) $\theta(f) \in \mathcal{D}^{\mathcal{G}}$. We show that those two conditions also hold for every $f \in \mathcal{R}^{\mathcal{G}}_{\alpha+1}$. As for (1), suppose that $f \in \mathcal{R}^{\mathcal{G}}_{\alpha+1}$ and, for some a, $\theta(f) = \theta(a)$. Now, $\theta(f) = \{\langle\theta(p), \theta(g)\rangle | \langle p, g\rangle \in f\}$ and $\theta(a) = \{\langle\theta(p), \theta(g)\rangle | \langle p, g\rangle \in a\}$. To show that $f = a$, we show that $f \subseteq a$ and $a \subseteq f$. We prove the first case, that is, $f \subseteq a$; the other case is similar. Suppose $x \in f$. Then $x = \langle p, a\rangle$ for some p and a. Then it follows that $\langle\theta(p), \theta(g)\rangle \in \theta(f)$. But, because we assumed $\theta(f) = \theta(a)$, it follows that $\langle\theta(p), \theta(g)\rangle \in \theta(a)$. Hence, by the definition of

$\theta(a)$, it follows that $\langle q, h\rangle \in a$ for some q, h where $\theta(p) = \theta(q)$ and $\theta(g) = \theta(h)$. But, by the induction assumption, the fact that $g \in \mathcal{R}^{\mathcal{G}}_{\alpha}$ and the fact that $\theta \restriction \mathcal{G}$ is 1-1, it follows that $p = q$ and $g = h$. Thus $\langle p, g\rangle = \langle q, h\rangle$, meaning $x \in a$.

Next we show that (2): for every $f \in \mathcal{R}^{\mathcal{G}}_{\alpha+1}$, $f \in \mathrm{Ran}(\theta)$. Now, all members of f are of the form $\langle p, g\rangle$ where $g \in \mathcal{R}^{\mathcal{G}}_{\alpha}$. Hence, it follows by the induction assumption that $g \in \mathrm{Ran}(\theta)$ for each such g. Define a set k as follows: $k = \{\langle q, h\rangle | \theta(q) = p \text{ and } \theta(h) = g \text{ for some } \langle p, g\rangle \in f\}$. Such a set exists by the induction assumption, and because k is a set, it follows that it must be a subset of $R^{\mathcal{G}}_{\beta}$ for some β, and consequently, it is a member of $R^{\mathcal{G}}_{\beta+1}$ (I was not fully sure of this, though it seemed to be true). Hence k is in the domain of θ, and because $\theta(k) = f$, it follows that $f \in \mathrm{Ran}(\theta)$.

Lastly, we show that (3): for every $f \in \mathcal{R}^{\mathcal{G}}_{\alpha+1}$, $\theta(f) \in \mathcal{D}^{\mathcal{G}}$. By definition, $\theta(f) = \{\langle \theta(p), \theta(g)\rangle | \langle p, g\rangle \in f\}$. By the induction assumption, and the fact that $\theta \restriction \mathcal{G}$ is 1-1 onto $\mathcal{G}$, it follows that $\theta(f) \subseteq R^{\mathcal{G}}_{\beta}$ for some β, and consequently, it is a member of $R^{\mathcal{G}}_{\beta+1}$ (again, I was not fully sure of this). Hence $\theta(f) \in \mathcal{D}^{\mathcal{G}}$.

The limit ordinal case is obvious.

□

Thus we know, by transfinite induction, that θ is 1-1, and is such that the range of θ includes all members and only members of $\mathcal{D}^{\mathcal{G}}$. Hence we know $\theta : \mathcal{D}^{\mathcal{G}} \xrightarrow[\text{onto}]{\text{1-1}} \mathcal{D}^{\mathcal{G}}$. The last two things we must show are that, if e is the identity automorphism of $\langle \mathcal{G}, \mathcal{R}\rangle$, then e is the identity map on $\mathcal{D}^{\mathcal{G}}$, and similarly for inverses.

First, suppose that e is the identity automorphism of $\langle \mathcal{G}, \mathcal{R}\rangle$. We prove, by transfinite induction, that $e(f) = f$ for all $f \in \mathcal{D}^{\mathcal{G}}$.

Proof. The least ordinal case is obvious, and likewise for the limit ordinal case.

Suppose the result holds for some α, that is, for all $f \in R^{\mathcal{G}}_{\alpha}$, $e(f) = f$. Consider any $f \in R^{\mathcal{G}}_{\alpha+1}$. Then, if $f \neq \emptyset$,

$$\begin{array}{rcll} e(f) & = & \{\langle e(p), e(g)\rangle | \langle p, g\rangle \in f\} & \\ & = & \{\langle p, g\rangle | \langle p, g\rangle \in f\} & \text{(By induction assumption and fact that } e \text{ is identity map)} \\ & = & f & \end{array}$$

On the other hand, if $f = \emptyset$, $e(f) = f$ is obvious, and this completes the proof.

□

Second, suppose that θ^{-1} is an inverse of some automorphism θ of $\langle \mathcal{G}, \mathcal{R}\rangle$. We show that θ^{-1} is the inverse of θ on $\mathcal{D}^{\mathcal{G}}$. We prove this by showing that, for all $f \in \mathcal{D}^{\mathcal{G}}$, $\theta\theta^{-1}(f) = f$ and $\theta^{-1}\theta = f$. We do this by using transfinite induction.

Proof. Again, the least ordinal and limit ordinal cases are obvious.

Suppose the result holds for some α, that is, for all $f \in R^{\mathcal{G}}_{\alpha}$, $\theta\theta^{-1}(f) = f$ and $\theta^{-1}\theta = f$. Consider any $f \in R^{\mathcal{G}}_{\alpha+1}$ (where $f \neq \emptyset$; the case where $f = \emptyset$ is obvious). We first show that $\theta\theta^{-1}f = f$. Now, $\theta^{-1}(f) = \{\langle \theta^{-1}(p), \theta^{-1}(g)\rangle | \langle p, g\rangle \in f\}$. Hence

$$\begin{array}{rl} & \theta\theta^{-1}(f) \\ = & \{\langle \theta\theta^{-1}(p), \theta\theta^{-1}(g)\rangle | \langle p, g\rangle \in f\} \\ = & \{\langle p, q\rangle | \langle p, g\rangle \in f\} \\ = & f \end{array}$$

by the induction assumption and the fact that θ^{-1} is the inverse of θ on $\langle \mathcal{G}, \mathcal{R}\rangle$. Thus we know that $\theta\theta^{-1}(f) = f$. Next we must show that $\theta^{-1}\theta = f$. Well, $\theta(f) = \{\langle \theta(p), \theta(g)\rangle | \langle p, g\rangle \in f\}$, and thus

$$\begin{array}{ll} & \theta^{-1}\theta(f) \\ = & \{\langle \theta^{-1}\theta(p), \theta^{-1}\theta(g)\rangle | \langle p, g\rangle \in f\} \\ = & \{\langle p, q\rangle | \langle p, g\rangle \in f\} \\ = & f \end{array}$$

by the induction assumption.

□

This completes the proof of Proposition 2.3.

Exercise 2.2

We prove that if a is an M-set (i.e., $a \in M$), $\theta(\hat{a}) = \hat{a}$ for every automorphism θ. Then we show that the counterparts of M-sets (i.e., $\hat{a}$ for each $a \in M$) are $\mathfrak{I}$-invariant for every automorphism sequence $\mathfrak{G}$. First we prove by transfinite induction (on the α for which $\hat{a} \in R^{\mathcal{G}}_{\alpha}$) that $\theta(\hat{a}) = \hat{a}$ for every automorphism θ and every M-set a.

Proof. Consider any automorphism θ. The least ordinal case is vacuous, and the limit ordinal case is obvious.

Suppose the result holds for some α. That is, suppose that for every $\hat{a} \in R^{\mathcal{G}}_{\alpha}$, $\theta(\hat{a}) = \hat{a}$. Consider any $\hat{b} \in R^{\mathcal{G}}_{\alpha+1}$. If $\hat{b} = \emptyset$, then the result obviously follows. So suppose $\hat{b} \neq \emptyset$. Then $\hat{b} = \{\langle p, \hat{g}\rangle | p \in \mathcal{G} \text{ and } g \in b\}$. Hence

$$\begin{array}{lll} & \theta(\hat{b}) = \{\langle \theta(p), \theta(\hat{g})\rangle | \langle p, g\rangle \in b\} & \\ = & \{\langle \theta(p), \hat{g}\rangle | \langle p, g\rangle \in b\} & \text{(By the induction assumption)} \\ = & \hat{b} & \text{(Because } \theta : \mathcal{G} \xrightarrow[\text{onto}]{\text{1-1}} \mathcal{G}\text{)} \end{array}$$

□

Now consider any automorphism sequence $\mathfrak{I}$ and some M-set a. Then, by the above result, $\theta(\hat{a}) = \hat{a}$ for every automorphism θ. Consequently $\hat{a}$ is $\mathfrak{G}$-invariant for every automorphism group $\mathfrak{G}$ of $\langle \mathcal{G}, \mathcal{R}\rangle$. To show that $\hat{a}$ is $\mathfrak{I}$-invariant, we must show that (1) for some $\mathfrak{G}$ in the sequence $\mathfrak{I}$, $\hat{a}$ is $\mathfrak{G}$-invariant; (2) for each $\langle p, g\rangle \in \hat{a}$, g is $\mathfrak{I}$-invariant. We prove these two things by induction on the $\mathcal{G}$-rank of $\hat{a}$.

Proof. Again, the least ordinal case is vacuous. Also, the case when $\hat{a} = \emptyset$ is likewise obvious, so suppose $\hat{a} \neq \emptyset$. Now suppose the result holds for all ordinals less than some ordinal α. We show that, for all $\hat{a}$ with $\mathcal{G}$-rank α, both (1) and (2) hold.

(1) This holds because, as remarked just before the start of this proof, $\hat{a}$ is $\mathfrak{G}$-invariant for *every* automorphism group $\mathfrak{G}$ of $\langle \mathcal{G}, \mathcal{R}\rangle$, hence also $\mathfrak{G}$-invariant for every $\mathfrak{G}$ in the sequence $\mathfrak{I}$.

(2) Consider any $\langle p, g\rangle \in \hat{a}$. By definition of $\hat{a}$, it follows that $g = \hat{b}$ for some b, and $\hat{b}$ must have $\mathcal{G}$-rank less than α. Thus, by the induction assumption, $\hat{b}$, that is, g, must be $\mathfrak{I}$-invariant.

□

This completes the proof of Proposition 2.9.

Section 3

Theorem 3.3

We show that Theorem 3.3 is provable from Lemma 3.4.

Theorem 0.1. Suppose $\mathfrak{J}$ is an automorphism sequence of $\langle \mathcal{G}, \mathcal{R} \rangle$ and $\theta \in \mathfrak{G}_0$. If f is $\mathfrak{J}$-invariant, so is $\theta(f)$.

Proof. By transfinite induction on the α for which $f \in R^{\mathcal{G}}_{\alpha}$. The least ordinal and limit ordinal cases are obvious. Suppose the result holds for some α. If $f = \emptyset$, then the result easily follows (because $\theta(f) = f$). So suppose $f \neq \emptyset$. Because f is $\mathfrak{J}$-invariant, it follows by definition that (1) for some $\mathfrak{G}_k$ in the sequence $\mathfrak{J}$, f is $\mathfrak{G}_k$-invariant; (2) for each $\langle p, g \rangle \in f$, g is $\mathfrak{J}$-invariant. To prove that $\theta(f)$ is $\mathfrak{J}$-invariant, we must show that (3) for some $\mathfrak{G}_j$ in the sequence $\mathfrak{J}$, f is $\mathfrak{G}_j$-invariant; (2) for each $\langle \theta(p), \theta(g) \rangle \in \theta(f)$ (for $\langle p, g \rangle \in f$), $\theta(g)$ is $\mathfrak{J}$-invariant. By Lemma 3.4, and the fact that f is $\mathfrak{G}_k$-invariant, $\theta(f)$ is also $\mathfrak{G}_k$-invariant, from which (3) follows immediately. As for (4), consider any $\langle \theta(p), \theta(g) \rangle \in \theta(f)$ where $\langle p, g \rangle \in f$. Then, by (2), g is $\mathfrak{J}$-invariant. But $g \in R^{\mathcal{G}}_{\alpha}$, and thus, by the induction assumption, $\theta(g)$ is also $\mathfrak{J}$-invariant. This completes the proof.

□

Theorem 3.5

We provide details of the proof of Theorem 3.5.

Theorem 0.2. Let X be any closed formula of $\mathcal{L}^*_C$ (the language with only $\in$ as a relation symbol) having all constants in $\mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$. For any $p \in \mathcal{G}$ and any $\theta \in \mathfrak{G}_0$, $p \vDash_{\mathfrak{J}} X$ iff $\theta(p) \vDash_{\mathfrak{J}} \theta(X)$.

Proof. First we note the things remarked in proof given of the theorem in the book. By the previous theorem, if X has all its constants in $\mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$, the same is true for $\theta(X)$, so $\theta(p) \vDash_{\mathfrak{J}} \theta(X)$ is a sensical statement. Now, the result is true for atomic X in $\mathcal{L}^*_C$, since at the atomic level, $\vDash_{\mathfrak{J}}$ and $\vDash$ agree, and we have Theorem 3.2. Now we prove the result for non-atomic formulas by induction on degree. This is done by considering the negation, disjunction, existential quantifier and the modal necessity cases.

Negation case: Suppose the result holds for X. We show it also holds for $\sim X$. That is, we show that $p \vDash_{\mathfrak{J}} \sim X$ iff $\theta(p) \vDash_{\mathfrak{J}} \theta(\sim X)$. On the one hand, suppose that $p \vDash_{\mathfrak{J}} \sim X$. Then $p \nvDash_{\mathfrak{J}} X$, meaning, by the induction assumption, $\theta(p) \nvDash_{\mathfrak{J}} \theta(X)$, hence $\theta(p) \vDash_{\mathfrak{J}} \sim\theta(X)$, which is the same as saying $\theta(p) \vDash_{\mathfrak{J}} \theta(\sim X)$. On the other hand, suppose that $\theta(p) \vDash_{\mathfrak{J}} \theta(\sim X)$. This is the same as saying $\theta(p) \vDash_{\mathfrak{J}} \sim\theta(X)$, hence $\theta(p) \nvDash_{\mathfrak{J}} \theta(X)$, meaning, by the induction assumption, $p \nvDash X$, hence $p \vDash \sim X$. Thus the negation case is complete.

Disjunction case: Suppose the result holds for X and Y. We show it also holds for $X \vee Y$. That is, we show that $p \vDash_{\mathfrak{J}} X \vee Y$ iff $\theta(p) \vDash_{\mathfrak{J}} \theta(X \vee Y)$. On the one hand, suppose that $p \vDash_{\mathfrak{J}} X \vee Y$. If $\theta(p) \nvDash_{\mathfrak{J}} \theta(X \vee Y)$, we would have $\theta(p) \nvDash_{\mathfrak{J}} \theta(X) \vee \theta(Y)$, hence both $\theta(p) \nvDash_{\mathfrak{J}} \theta(X)$ and $\theta(p) \nvDash_{\mathfrak{J}} \theta(Y)$, meaning, by the induction assumption, both $p \nvDash_{\mathfrak{J}} X$ and $p \nvDash_{\mathfrak{J}} Y$, meaning $p \vDash_{\mathfrak{J}} \sim(X \vee Y)$, a contradiction. Hence we must have $\theta(p) \vDash_{\mathfrak{J}} \theta(X \vee Y)$. On the other hand, suppose $\theta(p) \vDash_{\mathfrak{J}} \theta(X \vee Y)$. Then, if $p \nvDash X \vee Y$, we would have both $p \nvDash X$ and $p \nvDash Y$, meaning, by the induction assumption, both $\theta(p) \nvDash_{\mathfrak{J}} \theta(X)$ and $\theta(p) \nvDash_{\mathfrak{J}} \theta(Y)$, hence $\theta(p) \vDash_{\mathfrak{J}} \sim(\theta(X) \vee \theta(Y))$, that is, $\theta(p) \vDash_{\mathfrak{J}} \sim\theta(X \vee Y)$, a contradiction. Thus it must be that $p \vDash X \vee Y$. This concludes the disjunction case.

Existential quantifier case: Suppose, for all $a \in \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$, the result holds for $\phi(a)$ (for some $\phi(x)$ with x being the only free variable of ϕ). We must show that result also holds for $(\exists x)\phi(x)$. That is, we show that $p \vDash_{\mathfrak{J}} (\exists x)\phi(x)$ iff $\theta(p) \vDash_{\mathfrak{J}} \theta((\exists x)\phi(x))$. On the one hand, suppose $p \vDash_{\mathfrak{J}} (\exists x)\phi(x)$. Then, for some $a \in \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$, $p \vDash_{\mathfrak{J}} \phi(a)$. Then, by the induction assumption, $\theta(p) \vDash_{\mathfrak{J}} \theta(\phi(a))$. Let $\theta((\exists x)\phi(x)) = (\exists x)\psi(x)$. Now, $\theta(\phi(a)) = \psi(\theta(a))$, meaning $\theta(p) \vDash_{\mathfrak{J}} \psi(\theta(a))$, and thus $\theta(p) \vDash_{\mathfrak{J}} (\exists x)\psi(x)$, that is, $\theta(p) \vDash_{\mathfrak{J}} \theta((\exists x)\phi(x))$. On the other hand, suppose $\theta(p) \vDash_{\mathfrak{J}} \theta((\exists x)\phi(x))$. Again, let $\theta((\exists x)\phi(x)) = (\exists x)\psi(x)$. Thus we have $\theta(p) \vDash_{\mathfrak{J}} (\exists x)\psi(x)$, meaning $\theta(p) \vDash_{\mathfrak{J}} \psi(a)$ for some a. Now, $\psi(a) = \theta(\phi(\theta^{-1}(a)))$, meaning $\theta(p) \vDash_{\mathfrak{J}} \theta(\phi(\theta^{-1}(a)))$, and thus, by the induction assumption, $p \vDash_{\mathfrak{J}} \phi(\theta^{-1}(a))$, meaning $p \vDash_{\mathfrak{J}} (\exists x)\phi(x)$. This completes the existential quantifier case.

Modal necessity case: Suppose the result holds for X. We must show it also holds for $\Box X$. That is, we show that $p \vDash_{\mathfrak{J}} \Box X$ iff $\theta(p) \vDash_{\mathfrak{J}} \theta(\Box X)$, which is just saying $\theta(p) \vDash_{\mathfrak{J}} \Box\theta(X)$. On the one hand, suppose $p \vDash_{\mathfrak{J}} \Box X$. To show that $\theta(p) \vDash_{\mathfrak{J}} \Box\theta(X)$, consider any q with $\theta(p)\mathcal{R}q$. Then, because $\theta : \mathcal{G} \xrightarrow[\text{onto}]{\text{1-1}} \mathcal{G}$, $\theta(p') = q$ for some p'. Hence $\theta(p)\mathcal{R}\theta(p')$, meaning, because θ is an automorphism, $p\mathcal{R}p'$, and thus $p' \vDash_{\mathfrak{J}} X$, which implies $\theta(p') \vDash_{\mathfrak{J}} \theta(X)$ by the induction assumption, that is, $q \vDash_{\mathfrak{J}} \theta(X)$. Because q was arbitrary, it follows that $\theta(p) \vDash_{\mathfrak{J}} \Box\theta(X)$. On the other hand, suppose $\theta(p) \vDash_{\mathfrak{J}} \Box\theta(X)$. To show that $p \vDash_{\mathfrak{J}} \Box X$, consider any q with $p\mathcal{R}q$. Because θ is an automorphism, it follows that $\theta(p)\mathcal{R}\theta(q)$, and thus $\theta(q) \vDash_{\mathfrak{J}} \theta(X)$, meaning, by the induction assumption, $q \vDash_{\mathfrak{J}} X$. Because q was arbitrary, it follows that $p \vDash_{\mathfrak{J}} \Box X$. This concludes the modal necessity case, therefore completing the proof.

□

Section 4

Proposition 4.1

It is remakred in the second sentence of the first paragraph of the proof of Proposition 4.1 that $f \in g$ and $[\![f \in g]\!]$ are equivalent. This is not correct; $p \vDash f \in g$ implies $p \vDash [\![f \in g]\!]$, but $p \vDash [\![f \in g]\!]$ does not necessarily imply $p \vDash f \in g$. However, the atomic case is still simple as $p \vDash f \in g$ iff $p \vDash_{\mathfrak{J}} f \in g$ for every p (and $f, g \in \mathcal{D}_{\mathfrak{J}}^{\mathcal{G}}$), by the definition of $\vDash_{\mathfrak{J}}$. It follows that $p \vDash \Box\Diamond(f \in g)$ iff $p \vDash_{\mathfrak{J}} \Box\Diamond(f \in g)$ for every p, that is, $p \vDash [\![f \in g]\!]$ iff $p \vDash_{\mathfrak{J}} [\![f \in g]\!]$ for every p.

As a second note, we provide a proof of the second case mentioned, but omitted, in the proof given in the book.

Proof. That is, we suppose $\phi(x_1, ..., x_n)$ is $(\exists y)[y \in x_1 \wedge \psi(y, x_1, ..., x_n)]$, the result is known for ψ, $p \vDash_{\mathfrak{J}} [\![\phi(s_1, ..., s_n)]\!]$, where $s_1, ..., s_n \in \mathcal{D}_{\mathfrak{J}}^{\mathcal{G}}$, but $p \nvDash [\![\phi(s_1, ..., s_n)]\!]$. We derive a contradiction.

Because $p \nvDash [\![\phi(s_1, ..., s_n)]\!]$, $p' \vDash [\![\sim\phi(s_1, ..., s_n)]\!]$ for some p' with $p\mathcal{R}p'$, or equivalently, $p' \vDash [\![(\forall y)[y \in s_1 \supset \sim\psi(y, s_1, ..., s_n)]]\!]$.

Since $p \vDash_{\mathfrak{J}} [\![\phi(s_1, ..., s_n)]\!]$, also $p' \vDash_{\mathfrak{J}} [\![\phi(s_1, ..., s_n)]\!]$, that is, $p' \vDash_{\mathfrak{J}} [\![(\exists y)[y \in s_1 \wedge \psi(y, s_1, ..., s_n)]]\!]$. Then for some p'' with $p'\mathcal{R}p''$ and some $a \in \mathcal{D}_{\mathfrak{J}}^{\mathcal{G}}$, $p'' \vDash_{\mathfrak{J}} [\![[a \in s_1 \wedge \psi(a, s_1, ..., s_n)]]\!]$. It follows that both $p'' \vDash_{\mathfrak{J}} [\![(a \in s_1)]\!]$ and $p'' \vDash_{\mathfrak{J}} [\![\psi(a, s_1, ..., s_n)]\!]$. Because the atomic case has been established, $p'' \vDash [\![(a \in s_1)]\!]$. But also, since $p'\mathcal{R}p''$, $p'' \vDash [\![(\forall y)[y \in s_1 \supset \sim\psi(y, s_1, ..., s_n)]]\!]$, which implies $p'' \vDash [\![\sim\psi(a, s_1, ..., s_n)]\!]$. But $p'' \vDash_{\mathfrak{J}} [\![\psi(a, s_1, ..., s_n)]\!]$ has also been established, and this contradicts the induction hypothesis.

□

Exercise 4.1

(a) We suppose that X is Π-formula, and show that $\sim X$ is equivalent to some Σ-formula.

Proof. By induction on the complexity of X. The atomic case is obvious (because atomic formulas are Δ_0, hence both Σ and Π formulas). The different cases are for disjunction, conjunction, universal quantification, bounded universal quantification and bounded existential quantification.

Disjunction case: Suppose the result holds for two Π-formulas X and Y. We must show that it also holds for $X \vee Y$. Well, $\sim(X \vee Y)$ is equivalent to $\sim X \wedge \sim Y$. By the induction assumption, $\sim X$ and $\sim Y$ are equivalent to two Σ-formulas X' and Y', respectively. Hence $\sim(X \vee Y)$ is equivalent to $X' \wedge Y'$, which is a Σ-formula. This completes the disjunction case.

Conjunction case: Suppose the result holds for two Π-formulas X and Y. We must show that it also holds for $X \wedge Y$. Well, $\sim(X \wedge Y)$ is equivalent to $\sim X \vee \sim Y$. By the induction assumption, $\sim X$ and $\sim Y$ are equivalent to two Σ-formulas X' and Y', respectively. Hence $\sim(X \wedge Y)$ is equivalent to $X' \vee Y'$, which is a Σ-formula. This completes the conjunction case.

Universal quantification case: Suppose the result holds for some Π-formula ϕ. We must show it also holds for $(\forall x)\phi$. Well, $\sim(\forall x)\phi$ is equivalent to $(\exists x)\sim\phi$. By the induction assumption, $\sim\phi$ is equivalent to ψ for some Σ-formula ψ. Hence $\sim(\forall x)\phi$ is equivalent to $(\exists x)\psi$, which is a Σ-formula. This completes the universal quantification case.

Bounded universal quantification: Suppose the result holds for some Π-formula ϕ. We must show it also holds for $(\forall x \in y)\phi$. Well, $\sim(\forall x \in y)\phi$ is equivalent to $(\exists x \in y)\sim\phi$. By the induction assumption, $\sim\phi$ is equivalent to ψ for some Σ-formula ψ. Hence $\sim(\forall x \in y)\phi$ is equivalent to $(\exists x \in y)\psi$, which is a Σ-formula. This completes the bounded universal quantification case.

Bounded existential quantification: Suppose the result holds for some Π-formula ϕ. We must show it also holds for $(\exists x \in y)\phi$. Well, $\sim(\exists x \in y)\phi$ is equivalent to $(\forall x \in y)\sim\phi$. By the induction assumption, $\sim\phi$ is equivalent to ψ for some Σ-formula ψ. Hence $\sim(\exists x \in y)\phi$ is equivalent to $(\forall x \in y)\psi$, which is a Σ-formula. This completes the bounded existential quantification case, therefore completing the proof of (a).

□

(b) The proof of (b) is nearly identical to that of (a), so we omit it.

Exercise 4.2

We prove Corollary 4.3. That is, let $\phi(x_1, ..., x_n)$ be a formula of $\mathcal{L}_C$, and $s_1, ..., s_n$ be members of $\mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$. If ϕ is a Σ-formula and $p \vDash_{\mathfrak{J}} [\![\phi(s_1, ..., s_n)]\!]$, we show that $p \vDash [\![\phi(s_1, ..., s_n)]\!]$. If ϕ is a Π-formula and $p \vDash [\![\phi(s_1, ..., s_n)]\!]$, we show that $p \vDash_{\mathfrak{J}} [\![\phi(s_1, ..., s_n)]\!]$.

Proof. By induction on the complexity of ϕ.

(a) First, suppose that ϕ is a Σ-formula. Also suppose that $p \vDash_{\mathfrak{J}} [\![\phi(s_1, ..., s_n)]\!]$. We must show that $p \vDash [\![(s_1, ..., s_n)]\!]$. The case where ϕ is also a Δ_0 formula is immediate from Proposition 4.1. Thus we need only consider the disjunction, conjunction, existential quantifier, bounded universal quantifier and bounded existential quantifier cases.

Disjunction case: Suppose $\phi(s_1, ..., s_n)$ is of the form $\psi(s_1, ..., s_n) \vee \delta(s_1, ..., s_n)$, but $p \nvDash [\![\phi(s_1, ..., s_n)]\!]$. We derive a contradiction. Because $p \nvDash [\![\phi(s_1, ..., s_n)]\!]$, it follows that $p' \vDash [\![\sim\phi(s_1, ..., s_n)]\!]$ for some p' with $p\mathcal{R}p'$. Also, for some p'' with $p'\mathcal{R}p''$, at least one of $p'' \vDash_{\mathfrak{J}} [\![\psi(s_1, ..., s_n)]\!]$ and $p'' \vDash_{\mathfrak{J}} [\![\delta(s_1, ..., s_n)]\!]$ holds. Suppose, without loss of generality, that the former is the case, that is, $p'' \vDash_{\mathfrak{J}} [\![\psi(s_1, ..., s_n)]\!]$. By the induction assumption, this means $p'' \vDash [\![\psi(s_1, ..., s_n)]\!]$. Because $p'\mathcal{R}p''$, $p'' \vDash [\![\sim\phi(s_1, ..., s_n)]\!]$, and hence $p'' \nvDash [\![\phi(s_1, ..., s_n)]\!]$, meaning $p'' \nvDash [\![\psi(s_1, ..., s_n)]\!]$, a contradiction.

Conjunction case: Suppose $\phi(s_1, ..., s_n)$ is of the form $\psi(s_1, ..., s_n) \wedge \delta(s_1, ..., s_n)$, but $p \nvDash [\![\phi(s_1, ..., s_n)]\!]$. We derive a contradiction. Because $p \nvDash [\![\phi(s_1, ..., s_n)]\!]$, it follows that $p' \vDash [\![\sim\phi(s_1, ..., s_n)]\!]$ for some p' with $p\mathcal{R}p'$. Also, $p' \vDash_{\mathfrak{J}} [\![\psi(s_1, ..., s_n)]\!]$ and $p' \vDash_{\mathfrak{J}} [\![\delta(s_1, ..., s_n)]\!]$ both hold, meaning, by the induction assumption, both $p' \vDash [\![\psi(s_1, ..., s_n)]\!]$ and $p' \vDash [\![\delta(s_1, ..., s_n)]\!]$ hold. Because $p' \vDash [\![\sim\phi(s_1, ..., s_n)]\!]$, either $p' \nvDash [\![\psi(s_1, ..., s_n)]\!]$ or $p' \nvDash [\![\delta(s_1, ..., s_n)]\!]$, and this is a contradiction.

Existential quantifier case: Suppose $\phi(s_1, ..., s_n)$ is of the form $(\exists x)\phi(x, s_1, ..., s_n)$, but $p \nvDash [\![(\exists x)\phi(x, s_1, ..., s_n)]\!]$. We derive a contradiction. This means $p' \vDash [\![\sim(\exists x)\phi(x, s_1, ..., s_n)]\!]$ for some p' with $p\mathcal{R}p'$. Because $p \vDash_{\mathfrak{J}} [\![(\exists x)\phi(x, s_1, ..., s_n)]\!]$, it follows by the induction assumption that, for some $f \in \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$ and some p'' with $p'\mathcal{R}p''$, $p'' \vDash [\![\phi(f, s_1, ..., s_n)]\!]$. Then $p'' \vDash [\![(\exists x)(x, s_1, ...s_n)]\!]$. But we also have $p' \vDash [\![\sim(\exists x)\phi(x, s_1, ..., s_n)]\!]$, and this implies $p'' \vDash [\![\sim(\exists x)\phi(x, s_1, ..., s_n)]\!]$, a contradiction.

Bounded universal quantifier case: Suppose $\phi(s_1, ..., s_n)$ is of the form $(\forall x \in s_1)\psi(x, s_1, ..., s_n)$, but $p \nvDash [\![\phi(s_1, ..., s_n)]\!]$. We derive a contradiction. This means $p' \vDash [\![\sim(\forall x \in s_1)\phi(x, s_1, ..., s_n)]\!]$ for some p' with $p\mathcal{R}p'$, which implies $p'' \vDash [\![d \in s_1]\!]$ and $p'' \vDash [\![\sim\phi(d, s_1, ..., s_n)]\!]$ for some p'' with $p'\mathcal{R}p''$ and some $d \in \mathcal{D}^{\mathcal{G}}$. Because $p'' \vDash [\![d \in s_1]\!]$, it follows that $p''' \vDash [\![f \epsilon s_1]\!]$ and $p''' \vDash [\![d \approx f]\!]$ for some f and some p''' with $p''\mathcal{R}p'''$. Because $s_1 \in \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$, s_1 is $\mathfrak{J}$-invariant, which means, by Proposition 2.8, that g is $\mathfrak{J}$-invariant as well, and thus $g \in \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$. Then, by substitutivity, $p''' \vDash [\![\sim\phi(g, s_1, ..., s_n)]\!]$. Also, by Proposition 4.1, $p''' \vDash_{\mathfrak{J}} [\![g \in s_1]\!]$. But also $p''' \vDash_{\mathfrak{J}} [\![(\forall x \in s_1)\psi(x, s_1, ..., s_n)]\!]$ (because $p \vDash_{\mathfrak{J}} [\![(\forall x \in s_1)\psi(x, s_1, ..., s_n)]\!]$), which means $p''' \vDash_{\mathfrak{J}} [\![\psi(g, s_1, ..., s_n)]\!]$, hence $p''' \vDash [\![\psi(g, s_1, ..., s_n)]\!]$ by the induction hypothesis, which is a contradiction.

Bounded existential quantifier case: This is even easier than the bounded universal quantifier case, and we omit it.

(b) Next, suppose that ϕ is a Π-formula. Also suppose that $p \vDash [\![\phi(s_1, ..., s_n)]\!]$. We must show that $p \vDash_{\mathfrak{J}} [\![\phi(s_1, ..., s_n)]\!]$. The case where ϕ is also a Δ_0 formula is immediate from Proposition 4.1. The disjunction and conjunction cases are identical to those of part (a), while the bounded universal quantifier case is even easier than that of (a). Thus we only consider the universal quantifier and bounded existential quantifier cases.

Universal quantifier case: Suppose $\phi(s_1, ..., s_n)$ is of the form $(\forall x)\phi(x, s_1, ..., s_n)$, but $p \nvDash_{\mathfrak{J}} [\![(\forall x)\phi(x, s_1, ..., s_n)]\!]$. We derive a contradiction. This means $p' \vDash_{\mathfrak{J}} [\![\sim\phi(d, s_1, ..., s_n)]\!]$ for some $d \in \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$ and some p' with $p\mathcal{R}p'$. Because $p \vDash [\![(\forall x)\phi(x, s_1, ..., s_n)]\!]$, also $p' \vDash [\![(\forall x)\phi(x, s_1, ..., s_n)]\!]$, meaning $p' \vDash [\![\phi(d, s_1, ..., s_n)]\!]$. But by the induction hypothesis, this means $p' \vDash_{\mathfrak{J}} [\![\phi(d, s_1, ..., s_n)]\!]$, and this is a contradiction.

Bounded existential quantifier case: Suppose $\phi(s_1, ..., s_n)$ is of the form $(\exists x \in s_1)\phi(x, s_1, ..., s_n)$, but $p \nvDash_{\mathfrak{J}} [\![(\exists x \in s_1)\phi(x, s_1, ..., s_n)]\!]$. We derive a contradiction. Because $p \nvDash_{\mathfrak{J}} [\![(\exists x \in s_1)\phi(x, s_1, ..., s_n)]\!]$, it follows that $p' \vDash_{\mathfrak{J}} [\![\sim(\exists x \in s_1)\phi(x, s_1, ..., s_n)]\!]$ for some p' with $p\mathcal{R}p'$, that is, $p' \vDash_{\mathfrak{J}} [\![(\forall x \in s_1)\sim\phi(x, s_1, ..., s_n)]\!]$. Because $p \vDash [\![(\exists x \in s_1)\phi(x, s_1, ..., s_n)]\!]$, also $p' \vDash [\![(\exists x \in s_1)\phi(x, s_1, ..., s_n)]\!]$, and thus $p'' \vDash [\![d \in s_1]\!]$ and $p'' \vDash [\![\phi(d, s_1, ..., s_n)]\!]$ for some $d \in \mathcal{D}^{\mathcal{G}}$. Because $p'' \vDash [\![d \in s_1]\!]$, $p''' \vDash [\![d \approx f]\!]$ and $p''' \vDash [\![f \epsilon s_1]\!]$ both hold for some f. Using similar reasoning as in the bounded universal quantifier case of (a), it follows that $f \in \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$. By substitutivity, $p''' \vDash [\![\phi(f, s_1, ..., s_n)]\!]$, meaning $p''' \vDash_{\mathfrak{J}} [\![\phi(f, s_1, ..., s_n)]\!]$ by the induction hypothesis. But also $p''' \vDash_{\mathfrak{J}} [\![f \in s_1]\!]$ (by Proposition 4.1) and $p''' \vDash_{\mathfrak{J}} [\![(\forall x \in s_1)\sim\phi(x, s_1, ..., s_n)]\!]$ (because $p'\mathcal{R}p'''$), meaning $p''' \vDash [\![\sim\phi(f, s_1, ..., s_n)]\!]$, and this is a contradiction.

□

Section 5

Note that I encountered multiple difficulties with Propositions 5.3 and 5.4 of this section.

Notes

Note that, if f is $\mathfrak{G}$-invariant and $\langle p, g\rangle \in f$, it does *not* follow that g is also $\mathfrak{G}$-invariant. For it could be that, for some $\theta \in \mathfrak{G}$, $[\![f \approx \theta(g)]\!]$ is valid in $\mathcal{M}$ only because there is no world p where either $p \vDash [\![a \epsilon f]\!]$ or $p \vDash [\![a \epsilon \theta(g)]\!]$ hold for any a (meaning $[\![f \approx \theta(g)]\!]$ would trivially hold). This means that, if f is $\mathfrak{J}$-invariant and $\mathfrak{G}$-invariant for some $\mathfrak{G}$ in the sequence $\mathfrak{J}$, it follows that g (for $\langle p, g\rangle \in f$) is $\mathfrak{J}$-invariant, but it does not follow that g is $\mathfrak{G}$-invariant.

Lemma 5.1

I will note two things about the proof of this lemma. First, there is an error in the last paragraph of part (1). It is claimed that, because $q \vDash [\![c \approx x' \wedge c \epsilon a]\!]$ and $\theta^{-1} \in \mathfrak{G}$, Theorem 3.2 and the $\mathfrak{G}$-invariance of a means $\theta^{-1}(q) \vDash [\![\theta^{-1}(c) \approx \theta^{-1}(x') \wedge \theta^{-1}(c) \epsilon a]\!]$. Supposedly, the reasoning was that, because $q \vDash [\![c \approx x' \wedge c \epsilon a]\!]$, Theorem 3.2 obtains $\theta^{-1}(q) \vDash [\![\theta^{-1}(c) \approx \theta^{-1}(x') \wedge \theta^{-1}(c) \epsilon \theta^{-1}(a)]\!]$. Then, because a is $\mathfrak{G}$-invariant, $\theta^{-1}(q) \vDash [\![a \approx \theta^{-1}(a)]\!]$, and thus, by substitutivity, $\theta^{-1}(q) \vDash [\![\theta^{-1}(c) \approx \theta^{-1}(x') \wedge \theta^{-1}(c) \epsilon a]\!]$. But this is an incorrect conclusion to draw, as substitutivity only applies for formulas in the language $\mathcal{L}_C$, whereas the formula it was used on involves an instance of ϵ. In order to fix this problem, note that, because $q \vDash [\![c \epsilon a]\!]$, we also have $q \vDash [\![c \in a]\!]$. *Then* we have $q \vDash [\![c \approx x' \wedge c \in a]\!]$, for which we can use substitutivity and Theorem 3.2 to receive $\theta^{-1}(q) \vDash [\![\theta^{-1}(c) \approx \theta^{-1}(x') \wedge \theta^{-1}(c) \in a]\!]$. This does not yield $\theta^{-1}(q) \vDash \theta^{-1}(x') \epsilon b$ as in the proof, but it can easily be seen to yield $\theta^{-1}(q) \vDash [\![\theta^{-1}(x') \epsilon b]\!]$, which is what is actually needed.

The second thing I will note are the details for part (2) of the proof, where we show that $p \vDash [\![a \approx b]\!]$.

Proof. Suppose $[\![a \approx b]\!]$ was not true at p - we derive a contradiction. Then, by Proposition 2.12 of Chapter 17, either $p \nvDash [\![(\forall x)[(x \epsilon a) \supset (\exists y)(y \epsilon b \wedge y \approx x)]]\!]$ or $p \nvDash [\![(\forall x)[(x \epsilon b) \supset (\exists y)(y \epsilon a \wedge y \approx x)]]\!]$.

Case 1) $p \not\models [\![(\forall x)[(x \epsilon a) \supset (\exists y)(y \epsilon b \wedge y \approx x)]]\!]$. Then it follows that, for some q with $p\mathcal{R}q$ and some c, $q \models [\![c \epsilon a]\!]$ and $q \models [\![\sim(\exists y)(y \epsilon b \wedge y \approx c)]\!]$. Because $q \models [\![c \epsilon a]\!]$ and $q \models [\![a \subseteq f]\!]$, it follows that $q \models [\![c \in f]\!]$, meaning $r \models [\![c \approx d]\!]$ and $r \models [\![d \epsilon f]\!]$ for some d and some r with $q\mathcal{R}r$. Because $r \models [\![d \epsilon f]\!]$, it follows that, because f is $\mathfrak{J}$-invariant, so is d. Also, because $f \in R^{\mathcal{G}}_{\alpha+1}$ and $r \models [\![d \epsilon f]\!]$, it follows that $d \in R^{\mathcal{G}}_{\alpha}$. Hence we know that $d \in R^{\mathcal{J}}_{\alpha} \cap \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$. By the definition of b, it then follows that $r \models d \epsilon b$, which implies $r \models [\![d \epsilon b]\!]$. Because we also have $r \models [\![c \approx d]\!]$, this yields $r \models [\![(\exists y)(y \epsilon b \wedge y \approx c)]\!]$, contradicting $q \models [\![\sim(\exists y)(y \epsilon b \wedge y \approx c)]\!]$ (which implies $r \models [\![\sim(\exists y)(y \epsilon b \wedge y \approx c)]\!]$).

Case 2) $p \not\models [\![(\forall x)[(x \epsilon b) \supset (\exists y)(y \epsilon a \wedge y \approx x)]]\!]$. Then it follows that, for some q with $p\mathcal{R}q$ and some c, $q \models [\![c \epsilon b]\!]$ and $q \models [\![\sim(\exists y)(y \epsilon a \wedge y \approx c)]\!]$. Because $q \models [\![c \epsilon b]\!]$, this implies $r \models [\![(d \approx c \wedge d \epsilon a)]\!]$ for some r with $q\mathcal{R}r$ and some d. Hence $r \models [\![(\exists y)(y \epsilon a \wedge y \approx c)]\!]$, but this contradicts $q \models [\![\sim(\exists y)(y \epsilon a \wedge y \approx c)]\!]$.

□

Exercise 5.1

We complete the proof of Theorem 5.2.

Proof. That is, we show that g behaves like a powerset in $\mathcal{M}_{\mathfrak{J}}$. What this means is that we show

$$[\![(\forall x)(x \epsilon g \equiv x \subseteq f)]\!]$$

is valid in $\mathcal{M}_{\mathfrak{J}}$. Suppose, on the contrary, that $p \not\models_{\mathfrak{J}} [\![(\forall x)(x \epsilon g \equiv x \subseteq f)]\!]$ for some possible world p. We derive a contradiction. It follows that, for some $d \in \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$, $p \not\models_{\mathfrak{J}} [\![(d \epsilon g \equiv d \subseteq f)]\!]$. There are two cases: $p \not\models_{\mathfrak{J}} [\![(d \epsilon g \supset d \subseteq f)]\!]$ or $p \not\models_{\mathfrak{J}} [\![(d \subseteq f \supset d \epsilon g)]\!]$.

Case 1) $p \not\models_{\mathfrak{J}} [\![(d \epsilon g \supset d \subseteq f)]\!]$. Then it follows that $q \models_{\mathfrak{J}} [\![\sim(d \epsilon g \supset d \subseteq f)]\!]$ for some q with $p\mathcal{R}q$. This means that, for some r with $q\mathcal{R}r$, $r \models_{\mathfrak{J}} [\![d \epsilon g]\!]$ and $r \models_{\mathfrak{J}} [\![\sim(d \subseteq f)]\!]$. Because $r \models_{\mathfrak{J}} [\![d \epsilon g]\!]$, it follows by the definition of g, that, for some r' with $r\mathcal{R}r'$, $r' \models_{\mathfrak{J}} [\![d \subseteq f]\!]$. But, because $r\mathcal{R}r'$, also $r' \models_{\mathfrak{J}} [\![\sim(d \subseteq f)]\!]$, and this is a contradiction.

Case 2) $p \not\models_{\mathfrak{J}} [\![(d \subseteq f \supset d \epsilon g)]\!]$. Then it follows that $q \models_{\mathfrak{J}} [\![\sim(d \subseteq f \supset d \epsilon g)]\!]$ for some q with $p\mathcal{R}q$. This means that, for some r with $q\mathcal{R}r$, $r \models_{\mathfrak{J}} [\![d \subseteq f]\!]$ and $r \models_{\mathfrak{J}} [\![\sim(d \epsilon g)]\!]$. Because $r \models_{\mathfrak{J}} [\![d \subseteq f]\!]$ and $d \in \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$, $r \models_{\mathfrak{J}} d \epsilon g$ follows by the definition of g, and thus $r \models_{\mathfrak{J}} [\![d \epsilon g]\!]$. But this contradicts $r \models_{\mathfrak{J}} [\![\sim(d \epsilon g)]\!]$.

□

Proposition 5.3 and Exercise 5.3

It was not clear to me how Proposition 5.3 was any different from Theorem 2.7 of Chapter 17. $\mathcal{M}_{\mathfrak{J}}$ is an S4 model, so it would seem that that theorem carries over to $\mathcal{M}_{\mathfrak{J}}$. Thus it seems to me that Exercise 5.2 could be answered by just giving the same proof as in Theorem 2.7 of Chapter 17. It was not clear to me what was meant by "the translate of the extentionsionality axiom will be needed", as it was not needed for the proof of Theorem 2.7 of Chapter 17.

Exercise 5.3 is basically immediate from what we already know about the relationship between $\approx$ and $=$. Recall that $p \models [\![a \approx b]\!]$ iff $p \models [\![a = b]\!]$. So, if $a, b \in \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$ and $p \in \mathcal{G}$, $p \models [\![a \approx b]\!]$ iff $p \models [\![a = b]\!]$ iff $p \models_{\mathfrak{J}} [\![a = b]\!]$ (by Proposition 4.1).

Proposition 5.4

There is an error in the proof of this proposition (as well as the proof of Proposition 6.1 of Chapter 17). In particular, it is claimed that members of g are "by definition" members of $\mathcal{D}^{\mathcal{G}}$. However, members of g are in fact by definition *not* members of $\mathcal{D}^{\mathcal{G}}$. This is because members v of g are of the form $\langle q', b\rangle$ for some $q' \in \mathcal{G}$ and some $b \in \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$. But if $v \in \mathcal{D}^{\mathcal{G}}$, then $v \in \mathcal{R}^{\mathcal{G}}_{\alpha+1}$ for some α. Then $v \subseteq \mathcal{G} \times \mathcal{R}^{\mathcal{G}}_{\alpha}$. But obviously that cannot happen if $v = \langle q', b\rangle$. In fact, this is not so much of an error with the proof, as an errant tangent. For, all members of g are in $\mathcal{G} \times \mathcal{D}^{\mathcal{G}}$ (more specifically, $\mathcal{G} \times \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$), so $g \in \mathcal{D}^{\mathcal{G}}$ as well.

There is another error in the proof of this proposition. In paricular, the axiom of substitution in M is used on F for ψ - yet it was never verified that ψ was a function over F. Indeed, it seems that ψ is *not* a function over F. For, suppose that $\psi(u,v)$ for some $u \in F$. Thus $u = \langle q, a\rangle$ and $v = \langle q', b\rangle$ for some $a, b \in \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$ and some q and q' with $q\mathcal{R}q'$ where $q' \vDash_{\mathfrak{J}} [\![\phi(a,b)]\!]$. Now, it could very well be that $q'\mathcal{R}q''$ for some $q'' \neq q'$. If that was the case, it would follow that $q'' \vDash_{\mathfrak{J}} [\![\phi(a,b)]\!]$, which would mean $\psi(u,v')$ where $v' = \langle q'', b\rangle$ - so that $v' \neq v$. This error is simple to fix. Because $\mathcal{M}$ is well founded, the axiom of collection holds in $\mathcal{M}$, so that we can have the set g that we want, even though ψ may not be a function over F (this is how it worked in the proof of Proposition 6.1 of Chapter 17).

Another error in the proof of this proposition occurs in the second paragraph of pg. 278. It is claimed that, because f is $\mathfrak{G}$-invariant and $q \vDash [\![b \epsilon f]\!]$, we have $\theta^{-1}(q) \vDash [\![\theta^{-1}(b) \epsilon f]\!]$. The error that is occuring here is basically identical to the one that I described in the first paragraph of the section of these notes titled "Lemma 5.1". Here is a corrected proof that $[\![g^* \subseteq \theta(g^*)]\!]$ is valid in $\mathcal{M}$ for every $\theta \in \mathfrak{G}$.

Proof. Suppose $[\![g^* \subseteq \theta(g^*)]\!]$ were not valid in $\mathcal{M}$, for some $\theta \in \mathfrak{G}$. Then, in the usual way, there would be some $q \in \mathcal{G}$ and some $a \in \mathcal{D}^{\mathcal{G}}$ such that $q \vDash [\![a \in g^*]\!]$, $q \vDash [\![(a \epsilon g^*)]\!]$, and $q \vDash [\![\sim(a \in \theta(g^*))]\!]$. We derive a contradiction from this.

Since $q \vDash (a \epsilon g^*)$, $\langle p, a\rangle \in g$ for some $p \in \mathcal{G}$ with $p\mathcal{R}q$, and so for some $\langle p_0, b\rangle \in F$, $\psi(\langle p_0, g\rangle, \langle p, a\rangle)$ is true. Since $\langle p_0, b\rangle \in F$, $p_0 \vDash_{\mathfrak{J}} [\![(\forall x)(x \in f \supset (\exists! y)\phi(x,y)]\!]$ and $p_0 \vDash [\![b \epsilon f]\!]$. Since $\psi(\langle p_0, b\rangle, \langle p, a\rangle)$, $p_0 \mathcal{R} p$ and $p \vDash_{\mathfrak{J}} [\![\phi(b,a)]\!]$. It follows that $q \vDash_{\mathfrak{J}} [\![(\forall x)(x \in f \supset (\exists! y)\phi(x,y)]\!]$ and $q \vDash [\![b \epsilon f]\!]$.

By Theorem 3.5 and the fact that f is $\mathfrak{G}$-invariant, we have $\theta^{-1}(q) \vDash_{\mathfrak{J}} [\![(\forall x)(x \in f \supset (\exists! y)\phi(x,y)]\!]$. Also, by Theorem 3.2 and the fact that we have $q \vDash [\![b \epsilon f]\!]$, we have $\theta^{-1}(q) \vDash [\![\theta^{-1}(b) \in f]\!]$. Then it follows that, for some $\theta^{-1}(q')$ with $\theta^{-1}(q) \mathcal{R} \theta^{-1}(q')$, $\theta^{-1}(q') \vDash [\![\theta^{-1}(c) \epsilon f]\!]$ for some $\theta^{-1}(c) \in \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$, where $\theta^{-1}(q') \vDash [\![\theta^{-1}(b) \approx \theta^{-1}(c)]\!]$. Hence, by definition of F, $\langle \theta^{-1}(q'), \theta^{-1}(c)\rangle \in F$. So for some $\langle r, d\rangle \in g$, $\psi(\langle \theta^{-1}(q'), \theta^{-1}(c)\rangle, \langle r, d\rangle)$, so $\theta^{-1}(q') \mathcal{R} r$ and $r \vDash_{\mathfrak{J}} [\![\phi(\theta^{-1}(c), d)]\!]$. Because $\langle r, d\rangle \in g$, $r \vDash [\![d \epsilon g^*]\!]$, and it follows that $r \vDash_{\mathfrak{J}} [\![d \in g^*]\!]$. Using Theorem 3.5, we have $\theta(r) \vDash_{\mathfrak{J}} [\![\phi(c, \theta(d))]\!]$ and $\theta(r) \vDash_{\mathfrak{J}} [\![\theta(d) \in \theta(g^*)]\!]$. Also, because $\theta^{-1}(q) \mathcal{R} \theta^{-1}(q')$, $q \mathcal{R} q'$. Then, because $\theta^{-1}(q') \mathcal{R} r$, $q' \mathcal{R} \theta(r)$, hence also $q \mathcal{R} \theta(r)$. Because $q' \vDash [\![c \in f]\!]$, this means $\theta(r) \vDash [\![c \in f]\!]$, hence also $\theta(r)_{\mathfrak{J}} [\![c \in f]\!]$. Also $\theta(r) \vDash_{\mathfrak{J}} [\![(\forall x)[x \in f \supset (\exists! y)\phi(x,y)]]\!]$. Hence $\theta(r) \vDash_{\mathfrak{J}} [\![(\exists! y)\phi(c,y)]\!]$. Because $p \vDash_{\mathfrak{J}} [\![\phi(b,a)]\!]$, we also have $\theta(r) \vDash_{\mathfrak{J}} [\![\phi(b,a)]\!]$. Because $\theta^{-1}(q') \vDash [\![\theta^{-1}(b) \approx \theta^{-1}(c)]\!]$, $\theta^{-1}(q') \vDash [\![\theta^{-1}(b) = \theta^{-1}(c)]\!]$. Because $\theta^{-1}(q') \mathcal{R} r$, $\theta(r) \vDash [\![b = c]\!]$, meaning $\theta(r) \vDash_{\mathfrak{J}} [\![b = c]\!]$ (by Proposition 4.1 and the fact that $x = y$ is a Δ_0 formula). So, by substitutivity, $\theta(r) \vDash_{\mathfrak{J}} [\![\phi(c,a)]\!]$, meaning $\theta(r) \vDash_{\mathfrak{J}} [\![a = \theta(d)]\!]$. We know that $\theta(r) \vDash_{\mathfrak{J}} [\![\theta(d) \in \theta(g^*)]\!]$, so also $\theta(r) \vDash_{\mathfrak{J}} [\![a \in \theta(g^*)]\!]$, meaning $\theta(r) \vDash [\![a \in \theta(g^*)]\!]$, which contradicts $q \vDash [\![\sim(a \in \theta(g^*))]\!]$ (because $q \mathcal{R} \theta(r)$). This completes the proof.

□

There is a typo in the second sentence of the penultimate paragraph of this proof on pg. 277. In this sentence, "$\theta \in \mathcal{G}$" should be replaced with "$\theta \in \mathfrak{G}$".

Lastly, here is an explanation for why, if $p \vDash_{\mathfrak{J}} [\![(\forall x)[x \in f \supset (\exists! y)\phi(x,y)]]\!]$ then $p \vDash_{\mathfrak{J}} [\![(\forall x)[x \in f \supset (\exists y)(y \in g^* \wedge \phi(x,y))]]\!]$. Suppose $p \vDash_{\mathfrak{J}} [\![(\forall x)[x \in f \supset (\exists! y)\phi(x,y)]]\!]$ but $p \nvDash_{\mathfrak{J}} [\![(\forall x)[x \in f \supset (\exists y)(y \in g^* \wedge \phi(x,y))]]\!]$ - we derive a contradiction. Because $p \nvDash_{\mathfrak{J}} [\![(\forall x)[x \in f \supset (\exists y)(y \in g^* \wedge \phi(x,y))]]\!]$, it follows that, for some p' with $p\mathcal{R}p'$ and some $d \in \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$, $p' \vDash_{\mathfrak{J}} [\![(d \epsilon f)]\!]$ and $q \vDash_{\mathfrak{J}} [\![\sim(\exists y)(y \in g^* \wedge \phi(d,y))]\!]$. Then it follows, by the definition of F, that $\langle p', d\rangle \in F$. Hence, for some $v = \langle q, b\rangle$ with $v \in g$, $p' \mathcal{R} q$ and $b \in \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$, $\psi(\langle p', d\rangle, \langle q, b\rangle)$, meaning $q \vDash_{\mathfrak{J}} [\![\phi(d,b)]\!]$. Now, because $\langle q, b\rangle \in g$, $\langle q', b\rangle \in g^*$ for every q' with $q\mathcal{R}q'$. This means that $q \vDash_{\mathfrak{J}} [\![(b \in g^*)]\!]$. Hence we have both $q \vDash_{\mathfrak{J}} [\![(b \in g^*)]\!]$ and $q \vDash_{\mathfrak{J}} [\![\phi(d,b)]\!]$, meaning $q \vDash_{\mathfrak{J}} [\![(\exists y)(y \in g^* \wedge \phi(d,y))]\!]$. But this contradicts $q \vDash_{\mathfrak{J}} [\![\sim(\exists y)(y \in g^* \wedge \phi(d,y))]\!]$.

Translates of the Unorderd Pair and Union Axioms

In this section we provide proofs that the translates of the unorderd pair and union axioms are valid in $\mathcal{M}$. Consider $\mathcal{M}_{\mathfrak{J}}$ for any automorphism sequence $\mathfrak{J}$ of $\langle \mathcal{G}, \mathcal{R}\rangle$. First we handle the case for the union axiom.

Theorem 0.3. For any $f \in \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$ there is some $g \in \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$ such that

$$[\![(\forall x)(x \in g \equiv x \in \cup f)]\!]$$

is valid over $\mathcal{M}_{\mathfrak{J}}$, where we write $x \in \cup f$ as an abbreviation for $(\exists y)(y \in f \wedge x \in y)$.

Proof. Suppose $f \in R^{\mathcal{G}}_{\alpha+2}$ for some α. Define a set g as follows:

$$\langle p, a\rangle \in g \leftrightarrow p \in \mathcal{G} \text{ and } a \in \mathcal{R}^{\mathcal{G}}_{\alpha} \text{ and } a \text{ is } \mathfrak{J}\text{-invariant and } p \vDash [\![a \in \cup f]\!]$$

It follows that $g \in R^{\mathcal{G}}_{\alpha+1}$, that is, $g \in \mathcal{D}^{\mathcal{G}}$. Also, it can easily be seen that if $p \vDash a \epsilon g$, then $p \vDash [\![a \epsilon g]\!]$ as well. We first verify that $g \in \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$. To show this, we must show that (1) g is $\mathfrak{G}$-invariant for some $\mathfrak{G}$ in the sequence $\mathfrak{J}$; and (2) for each $\langle p, a\rangle \in g$, a is $\mathfrak{J}$-invaraint. (2) follows by definition of g, so it is (1) that we must show.

Because $f \in \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$, f is $\mathfrak{G}$-invariant for some $\mathfrak{G}$ in the sequence $\mathfrak{J}$. We show that g is also $\mathfrak{G}$-invariant. In order to do this, we need to show that $[\![g \approx \theta(g)]\!]$ is valid in $\mathcal{M}$ for each $\theta \in \mathfrak{G}$. It suffices to show that $[\![g \subseteq \theta(g)]\!]$ for each $\theta \in \mathfrak{G}$. This is because, if we know that, then, because θ is closed under inverses, we would have $[\![g \subseteq \theta^{-1}(g)]\!]$ being valid in $\mathcal{M}$, hence also $[\![\theta(g) \subseteq g]\!]$ (by Theorem 3.2 and the fact that $\theta : \mathcal{G} \xrightarrow[onto]{1\text{-}1} \mathcal{G}$). Hence we only show that $[\![g \subseteq \theta(g)]\!]$ is valid in $\mathcal{M}$ for each $\theta \in \mathfrak{G}$.

Suppose, on the contrary, that $p \nvDash [\![g \subseteq \theta(g)]\!]$ for some p and some $\theta \in \mathfrak{G}$. Then, by the definition of g, it would follow that for some q with $p\mathcal{R}q$ and some $a \in \mathcal{D}^{\mathcal{G}}$, all of $q \vDash [\![a \epsilon g]\!]$, $q \vDash a \epsilon g$ and $q \vDash [\![\sim(a \in \theta(g))]\!]$ would hold. Because $q \vDash [\![a \epsilon g]\!]$, we also have $q \vDash [\![a \in g]\!]$. Also, because $q \vDash a \epsilon g$, we have $q \vDash [\![a \in \cup f]\!]$. Then, using the fact that f is $\mathfrak{G}$-invariant, we have $\theta^{-1}(q) \vDash [\![\theta^{-1}(a) \in \cup f]\!]$. Because f is $\mathfrak{G}$-invariant, and $a \in \cup f$ is an abbreviation for $(\exists y)(y \in f \wedge a \in y)$, it can then easily be seen to follow that, for some $b \in \mathcal{R}^{\mathcal{G}}_{\alpha+1} \cap \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$, some $c \in \mathcal{R}^{\mathcal{G}}_{\alpha} \cap \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$ and some $\theta^{-1}(q')$ with $\theta^{-1}(q)\mathcal{R}\theta^{-1}(q')$, $\theta^{-1}(q') \vDash [\![\theta^{-1}(a) \approx c]\!]$ and $\theta^{-1}(q') \vDash [\![b \in f \wedge c \in b]\!]$ both hold. By definition of g, this means that $\theta^{-1}(q') \vDash c \epsilon g$, hence $\theta^{-1}(q') \vDash [\![c \epsilon g]\!]$, so by Theorem 3.2, $q' \vDash [\![\theta(c) \in \theta(g)]\!]$. Then, because $q' \vDash [\![a \approx \theta(c)]\!]$ (because $\theta^{-1}(q') \vDash [\![\theta^{-1}(a) \approx c]\!]$), we have $q' \vDash [\![a \in \theta(g)]\!]$. Because θ^{-1} is an automorphism and $\theta^{-1}(q)\mathcal{R}\theta^{-1}(q')$, $q\mathcal{R}q'$. Hence $q' \vDash [\![\sim(a \in \theta(g))]\!]$ also holds (because $q \vDash [\![\sim(a \in \theta(g))]\!]$ and $q\mathcal{R}q'$), which is a contradiction.

Now that we know that g is $\mathfrak{G}$-invariant, we show that $[\![(\forall x)(x \in g \equiv x \in \cup f)]\!]$ is valid over $\mathcal{M}_{\mathfrak{J}}$. Suppose, on the contrary, that $p \nvDash_{\mathfrak{J}} [\![(\forall x)(x \in g \equiv x \in \cup f)]\!]$ held for some p - we derive a contradiction. It follows that $p \nvDash_{\mathfrak{J}} [\![(a \in g \equiv a \in \cup f)]\!]$ for some $a \in \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$. There are two cases.

Case 1) $p \nvDash_{\mathfrak{J}} [\![(a \in g \supset a \in \cup f)]\!]$. Then it follows that, for some q with $p\mathcal{R}q$, $q \vDash_{\mathfrak{J}} [\![a \in g]\!]$ and $q \vDash_{\mathfrak{J}} [\![\sim(a \in \cup f)]\!]$. It follows that for some r with $q\mathcal{R}r$ and some $b \in \mathcal{D}^{\mathcal{G}}_{\mathfrak{J}}$, $r \vDash_{\mathfrak{J}} [\![a \approx b]\!]$ and $r \vDash_{\mathfrak{J}} b \epsilon g$. By the definition of g, it follows that $r \vDash_{\mathfrak{J}} [\![b \in \cup f]\!]$. Because $q\mathcal{R}r$, we also have $r \vDash_{\mathfrak{J}} [\![\sim(a \in \cup f)]\!]$. By substitutivity, $r \vDash_{\mathfrak{J}} [\![\sim(b \in \cup f)]\!]$. But this contradicts $r \vDash_{\mathfrak{J}} [\![b \in \cup f]\!]$.

Case 2) $p \nvDash_{\mathfrak{J}} [\![(a \in \cup f \supset a \in g)]\!]$. Then it follows that, for some q with $p\mathcal{R}q$, $q \vDash_{\mathfrak{J}} [\![a \in \cup f]\!]$ and $q \vDash_{\mathfrak{J}} [\![\sim(a \in g)]\!]$. Because $q \vDash_{\mathfrak{J}} [\![a \in \cup f]\!]$, there is some r with $q\mathcal{R}r$ and some b such that $r \vDash_{\mathfrak{J}} [\![b \in f \wedge a \in b]\!]$. Because $f \in \mathcal{R}^{\mathcal{G}}_{\alpha+2}$, it follows that $r' \vDash_{\mathfrak{J}} [\![c \in f \wedge a \in c]\!]$ for some r' with $r\mathcal{R}r'$ and some $c \in \mathcal{R}^{\mathcal{G}}_{\alpha+1}$. Likewise, because $c \in \mathcal{R}^{\mathcal{G}}_{\alpha+1}$, it follows that $r'' \vDash_{\mathfrak{J}} [\![c \in f \wedge d \in c]\!]$ for some r'' with $r'\mathcal{R}r''$ and some $d \in \mathcal{R}^{\mathcal{G}}_{\alpha}$ where $r'' \vDash_{\mathfrak{J}} [\![a \approx d]\!]$. By the definition of g, it follows that $r'' \vDash_{\mathfrak{J}} d \epsilon g$, hence $r'' \vDash_{\mathfrak{J}} [\![d \epsilon g]\!]$, hence $r'' \vDash_{\mathfrak{J}} [\![d \in g]\!]$. But because $q\mathcal{R}r''$ we have $r'' \vDash_{\mathfrak{J}} [\![\sim(a \in g)]\!]$, meaning, by substitutivity, $r'' \vDash_{\mathfrak{J}} [\![\sim(d \in g)]\!]$, which is a contradiction.

□